HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · OIL & GAS · PIPELINES

ASSET INTEGRITY · INLINE INSPECTION TO DIG PLAN

Pipeline Defect Analytics

Thousands of inspection features became a 3D defect twin, every metal-loss anomaly got a standard fitness-for-service check, repeat runs gave each defect a growth rate, and digs were ranked by risk, with GenAI drafting the field briefs.

~~6~~ wk → 3 d
from inspection data to a dig plan (approx.)

3 days
From inspection data to a dig plan
▼ from about 6 weeks

100 %
Metal-loss anomalies assessed
every feature, not a sample

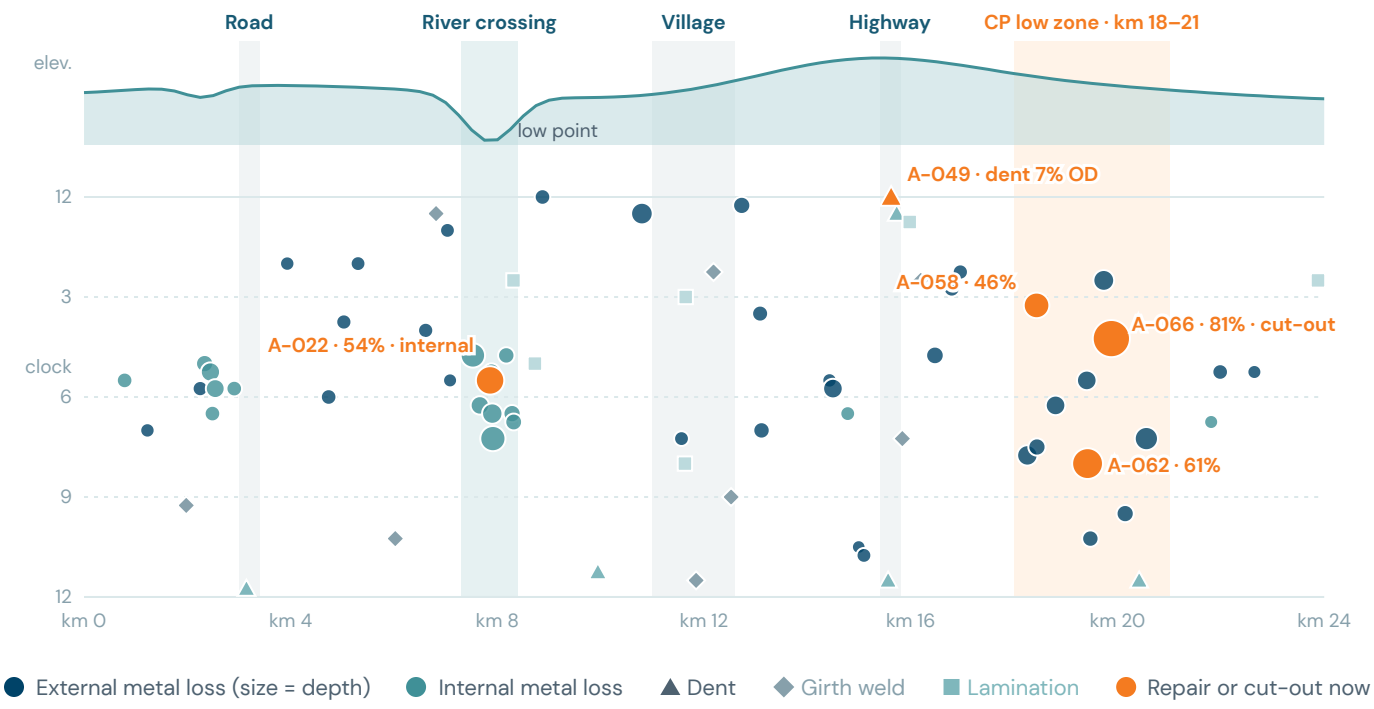
35 % fewer
Unnecessary excavations
▼ digs ranked by risk and growth

70 % lower
Engineering hours per inspection run
▼ automated assessment and briefs

Rings: 3 days against roughly 6 weeks (full inner ring); one segment per metal-loss feature in the demo segment (53 of 53 assessed); share of excavations avoided; hours still needed against the old 100%. Figures indicative; validated metrics available on request.

01 Every anomaly, in its place on the line

Unrolled wall map of a 24 km, 12-inch segment · 73 reported anomalies



Synthetic pipeline and inspection data from the page's demo; defect positions and sizes are illustrative. Elevation strip above; clock position below (12 = top of pipe, 6 = bottom of line).

HOTSPOT · EXTERNAL
CP low zone, km 18–21
12 external defects growing at up to 3.6% of wall a year: one 81% cut-out and two repairs. Cathodic protection needs attention.

HOTSPOT · INTERNAL
River crossing low point
9 internal defects at the bottom of the line, consistent with water drop-out. A-022 (54%) needs repair; more cleaning pigging and an inhibitor review.

CONTEXT
Dent at the highway crossing
A-049, 7% of OD under a crossing that scores 4 of 5 on consequence: repair.

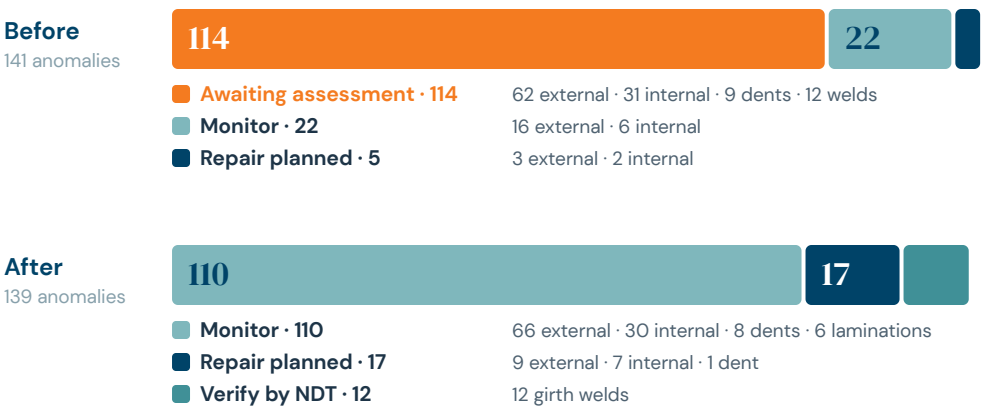
THE RUN IN ONE LINE
1 cut-out · 4 repair · 8 monitor
8 welds to verify by NDT, 52 acceptable. 15 digs needed within 5 years; immediate actions at 80% of MAOP.

02 No backlog, and corrosion slowing where it was worst

Impact dashboard · before vs after

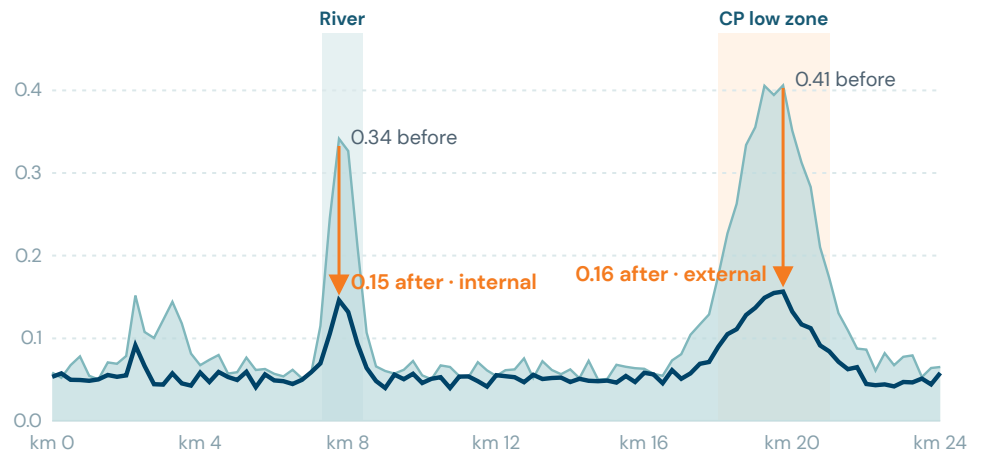
Anomaly backlog 30 days after a run

Anomalies by decision status, with the type split



Corrosion rate along the line

Before (light area) vs after (navy line), once cathodic protection and pigging fixes targeted the hot stretches



mm per year, faster of external or internal corrosion at each point

03 Four pain points, four fixes

Before → after

BEFORE
Feature lists, not insight
Each run arrived as vendor spreadsheets of depth, length and clock position, reviewed line by line.

3D DEFECT TWIN

AFTER
See the whole run
3D twin and unrolled wall map, from feature to field in one workspace.

BEFORE
Sampled assessments
Checks run by hand on the deepest features, so long shallow defects slipped through.

MODIFIED B31G ENGINE

AFTER
Every feature checked
Every metal-loss feature assessed with Modified B31G (0.85 dL) and location-class design factors.

BEFORE
Growth unknown
Repeat runs were not matched, so corrosion rates and remaining life were guesses.

RUN-TO-RUN MATCHING

AFTER
A rate per defect
2021 and 2026 runs matched; depth becomes remaining life, and life a dig calendar.

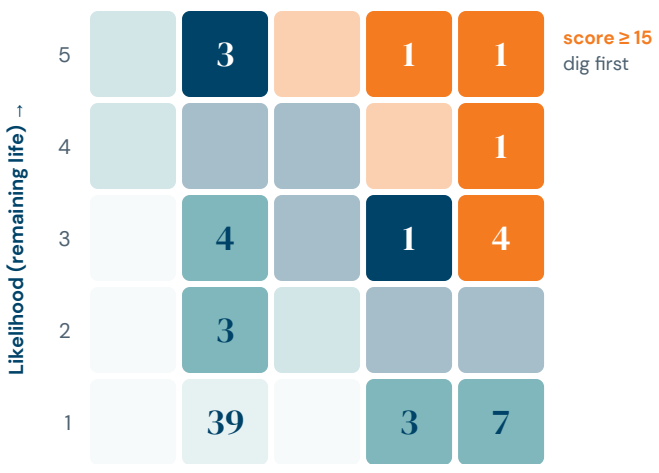
BEFORE
Context missing
Crossings, villages, CP readings and design factors sat apart from the defects; digs ranked on depth only.

GIS + RISK SCORE

AFTER
Risk, not depth
Remaining life × location consequence ranks the digs; GenAI drafts the brief in minutes.

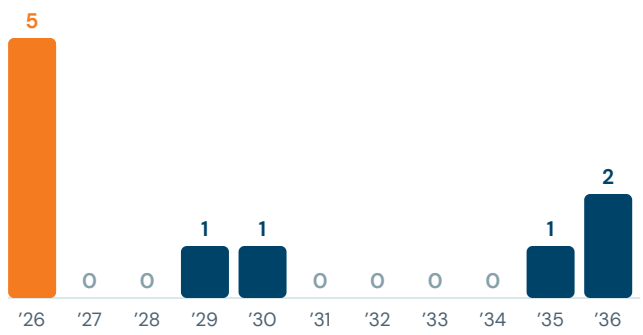
04 Where to dig first

Risk matrix, 2026 · defects per cell (laminations excluded)



05 The dig calendar

New digs due per year from projected growth



5 act-now features in 2026: 1 cut-out, 4 repairs.
Then a thin, plannable trickle to 2036.

06 How HMC ran it

2 weeks	**Discover** Worked past runs with the integrity team; vendor formats mapped. Data model and rulebook.
3 weeks	**Design** Standard methods, no black boxes: B31G, run matching, risk score. Engine spec.
6 weeks	**Build** 3D twin, assessment lab, dig planner, GenAI briefs and Power BI views.
1 segment	**Adopt** Pilot checked against manual assessments; engineers sign off every repair.
All segments	**Scale** New runs assessed within days; dig findings recalibrate growth rates.

What any pipeline operator can borrow

01
Assess every feature
Automating a standard method removes sampling bias; long, shallow defects get the same scrutiny as deep ones.

02
Growth beats a snapshot
Matched runs turn depth into remaining life, and remaining life into a dig calendar.

03
Context sets the priority
The same defect under a village or a river matters more than in an open field. Put location in the risk score.

Built with **Python** · **Three.js** · **D3** · **GIS layers** · **Large language model** · **Power BI** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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